

interface over HTTP. The following can be the REST API for adding a user:

POST /user

This API accepts the user data in the format of *application/json* in the HTTP request, and returns the newly created user data in the *application/json* format along with usual HTTP headers.

Let us do this one step at a time.

Generate the Spring Boot project

Visit the <https://start.spring.io/> and choose options, as shown in Figure 4. The following are some of the important options selected.

Project type: Maven

Language: Java

Spring Boot version: 2.7.0

Project metadata:

Group: *com.glarimy*

Artifact: *ums-add-service*

Package name: *com.glarimy.ums*

Packaging: JAR

Java version: 8

Dependencies: Spring Web, Rest Repositories, Spring Data JPA, H2 Database, Spring Boot Dev Tools, and Validation

Click on the *Generate* button and you will get the Maven project downloaded as an archive. Extract it and export the project into any IDE of your choice.

Manifest

As you know, the Maven tool uses the manifest file *pom.xml* to curate the build process. The following part is self-explanatory.

```
<parent>
  <groupId>org.springframework.boot</groupId>
  <artifactId>spring-boot-starter-parent</artifactId>
  <version>2.7.0-SNAPSHOT</version>
  <relativePath/>
</parent>
<groupId>com.glarimy</groupId>
```

The screenshot shows the Spring Initializr web application interface. It is divided into several sections for configuring a new project:

- Project:** Maven Project (selected), Gradle Project.
- Language:** Java (selected), Kotlin, Groovy.
- Spring Boot:** 2.7.0 (SNAPSHOT) (selected), 3.0.0 (M1), 2.7.0 (M2), 2.6.5 (SNAPSHOT), 2.6.4, 2.5.11 (SNAPSHOT), 2.5.10.
- Project Metadata:**
 - Group: com.glarimy
 - Artifact: user-add-service
 - Name: user-add-service
 - Description: Add Service for User Management
 - Package name: com.glarimy.ums
 - Packaging: Jar (selected), War.
 - Java: 17, 11, 8 (selected).
- Dependencies:** A list of selected dependencies with their descriptions:
 - Spring Web (WEB):** Build web, including RESTful, applications using Spring MVC. Uses Apache Tomcat as the default embedded container.
 - Rest Repositories (WEB):** Exposing Spring Data repositories over REST via Spring Data REST.
 - Spring Data JPA (SQL):** Persist data in SQL stores with Java Persistence API using Spring Data and Hibernate.
 - H2 Database (SQL):** Provides a fast in-memory database that supports JDBC API and R2DBC access, with a small (2mb) footprint. Supports embedded and server modes as well as a browser based console application.
 - Spring Boot DevTools (DEVELOPER TOOLS):** Provides fast application restarts, LiveReload, and configurations for enhanced development experience.
 - Validation (I/O):** Bean Validation with Hibernate validator.

At the bottom, there are three buttons: **GENERATE** (with a download icon), **EXPLORE** (with a keyboard shortcut CTRL + SPACE), and **SHARE...**

Figure 4: Spring Boot initialisation wizard